화학특강 (2010년 2학기)
소개하기

황현석

강원대학교 화학과

2010년 9월 6일 월요일

화학특강 (2010년 2학기) 강의계획서

- 담당교수: 황현석 (연구실: 자연대 1호관 313호, Email: hhwang@kangwon.ac.kr)
- 강의시간 및 강의실: 월, 목 4:30 - 6:00 PM, 자연대 1호관 407호
- 면담시간: 매주 수요일 3:00 - 4:00 PM
- 강의 홈페이지: http://www.kangwon.ac.kr/~cclab
- 시험계획 및 성적평가: **중간고사 1회 (35 %)+ 기말고사 1회 (35 %) + 발표 (30 %)**
- 강의 주교재 및 보조교재
  - 주교재: 강의 슬라이드
  - 보조교재 1: 정원양 외 역, “일반화학”, 자유아카데미 (2007).
  - 보조교재 2: P. Atkins & J. de Paula, “Physical Chemistry (8th. Ed.)”, Oxford University Press, New York (2006).

강의 목표와 내용

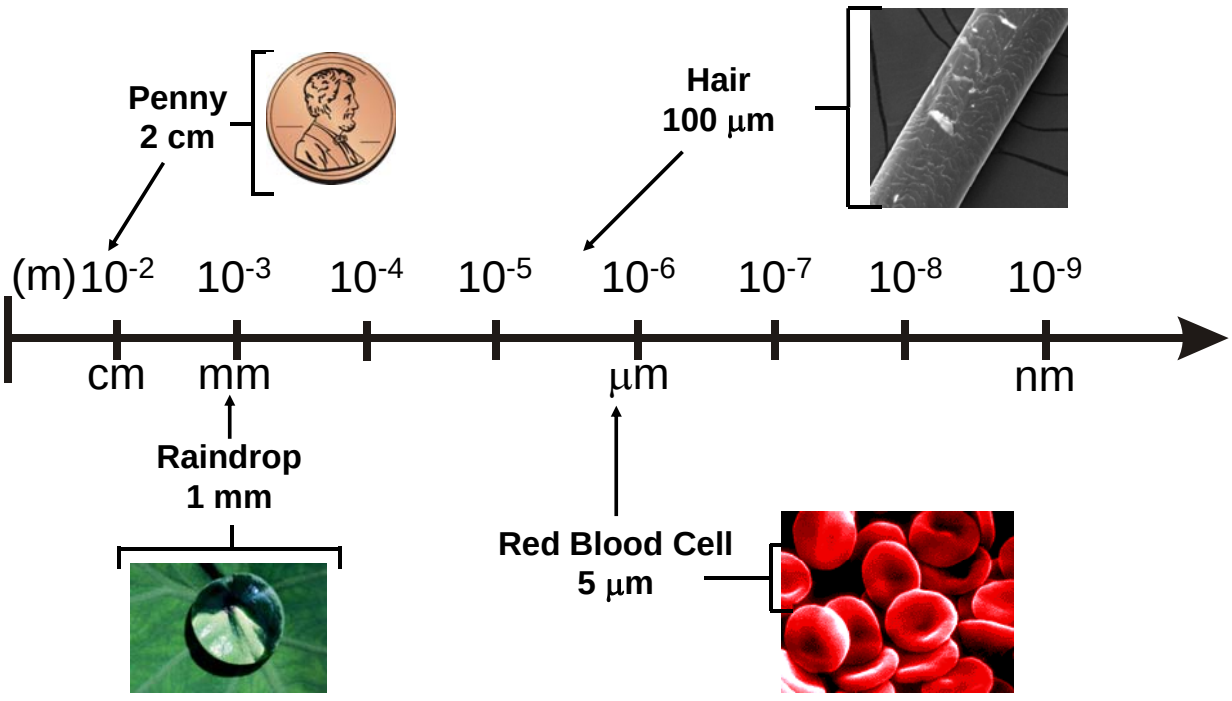
강의 목표:

1. 학부 과정에서 배운 화학 지식을 바탕으로 그 응용과 화학의 최근 발전 동향에 대해 알아본다.
2. 습득한 화학 지식을 체계적으로 정리하여 발표할 수 있는 능력을 키운다.

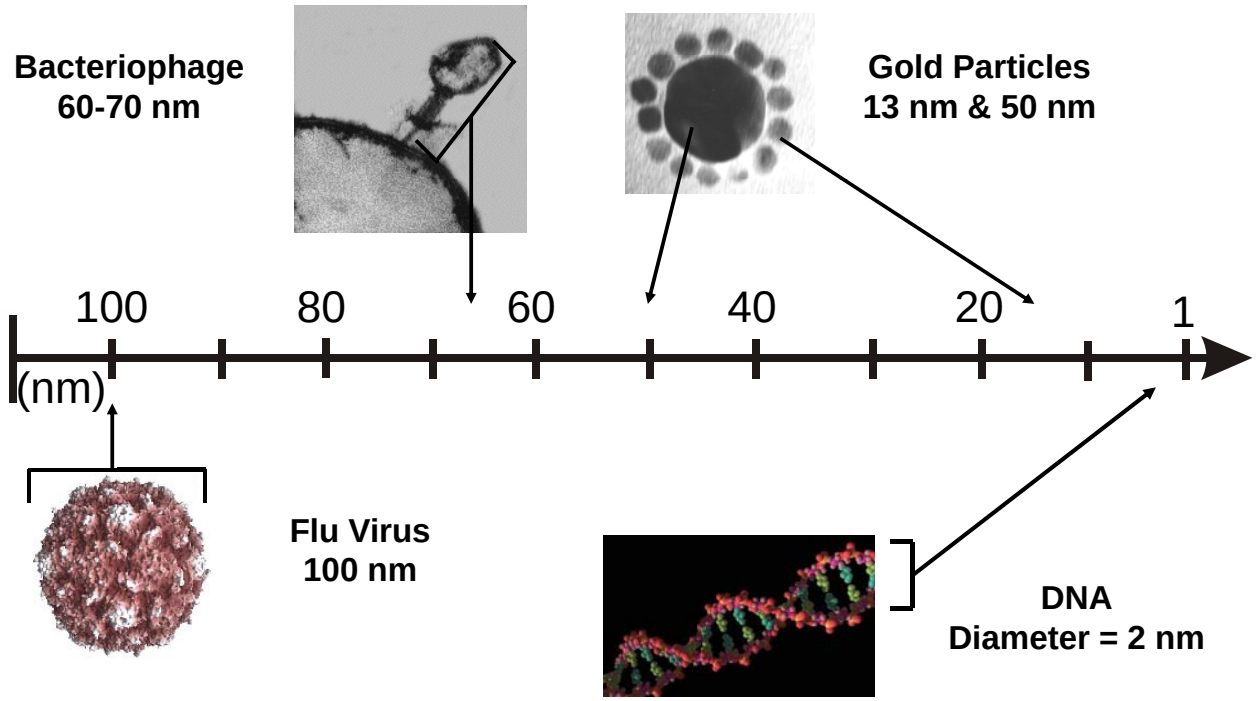
강의 내용

• 제 0 장	화학의 최근 발전 동향에 대한 소개 – Biotechnology & Nanotechnology 중심으로	중간고사
• 제 1 장	화학의 기본 개념 정리	
• 제 2 장	DNA와 Biotechnology	
• 제 3 장	Self-Assembly 와 그 응용	
• 제 4 장	고분자, Dendrimer 그리고 Supramolecular Chemistry	
• 제 5 장	Nanotechnology와 Quantum Dots	
• 제 6 장	신물질의 개발과 그 응용	
• 제 7 장	전기화학과 그 응용: 태양 전지와 연료 전지	
	발표 수업	기말고사

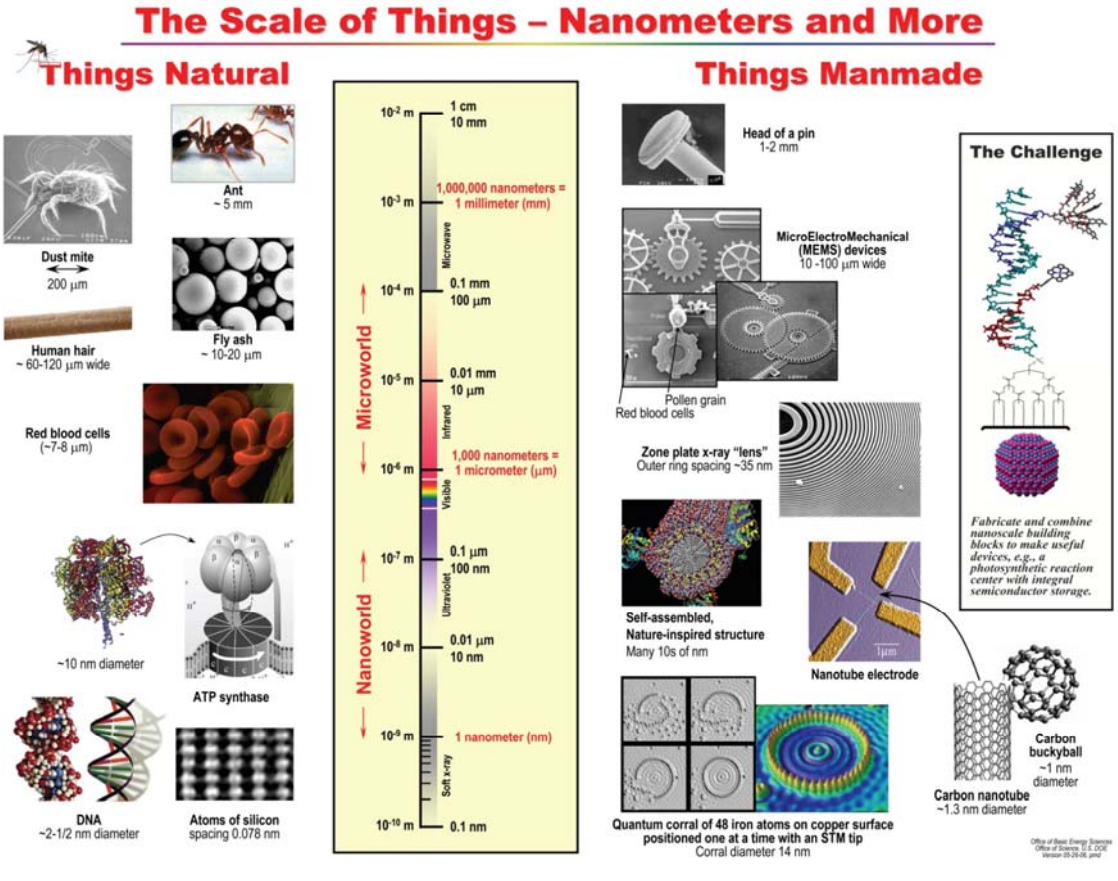
What Is A Nanometer?



Now Entering The NANO-ZONE



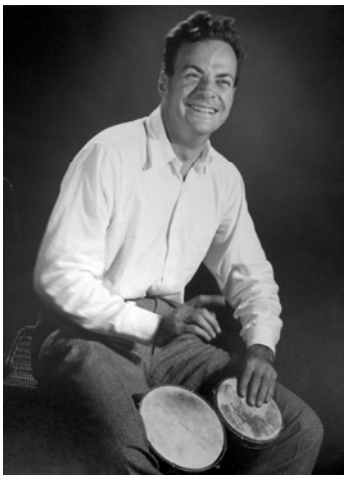
What is Nanotechnology?



What is Nanotechnology?



Origin of the term “Nanotechnology”



“There’s Plenty of Room at the Bottom”, the title of a talk given by physicist Richard Feynman



“**Nanotechnology** mainly consists of the processing of separation, consolidation, and deformation of materials by one atom or by one molecule.”
Norio Taniguchi, “On the Basic Concept of Nanotechnology”, *Proc. Intl. Conf. Prod.* London, Part II, 1974.

Nanotechnology Is Not a New Phenomenon

The Lycurgus Cup: 4th Century A.D.



Green = Reflected Light



Red = Transmitted Light

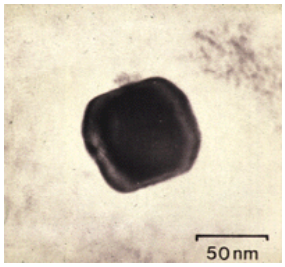


Image of silver/gold nanoparticle in the Lycurgus cup (300 ppmsilver, 40 ppm gold)

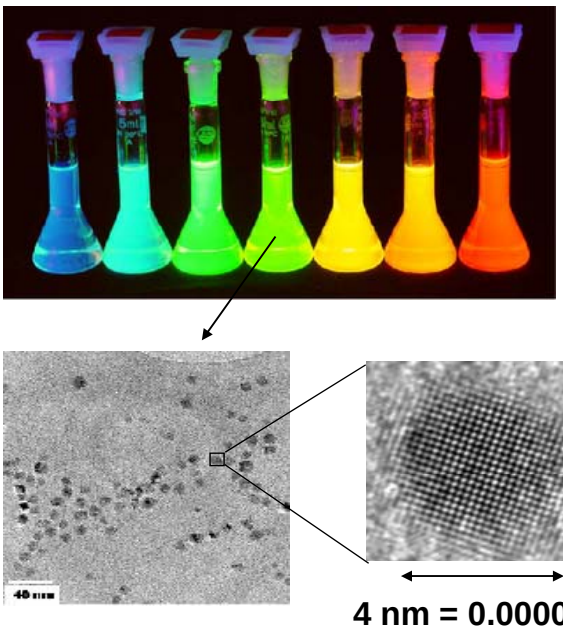
Nanotechnology Is Not a New Phenomenon

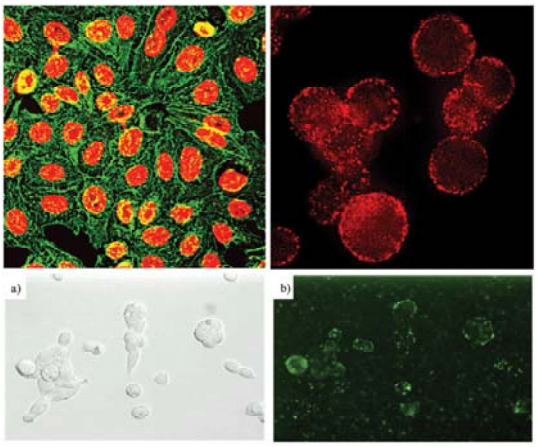


Stained glass windows in medieval churches

Quantum Dots and Their Applications

Solutions of monodisperse CdSe nanocrystals with increasing diameter (2 to 7 nm) from left to right



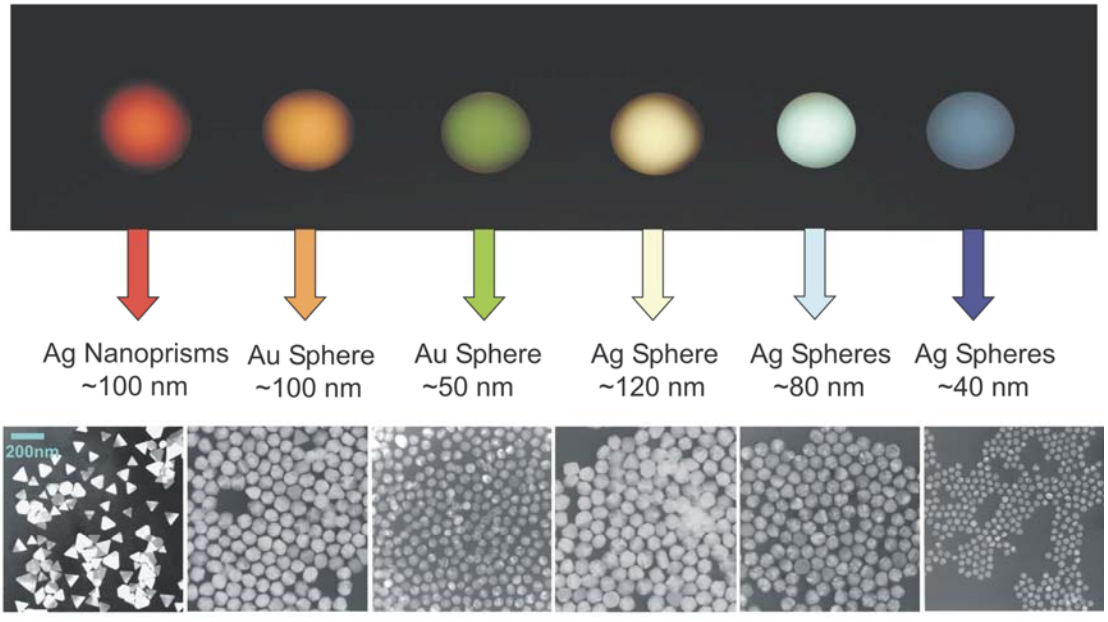


Parak et al., Nanotechnology **16**, R9 (2005)

4 nm = 0.000004 mm

Colour changes with Size!

Rayleigh Light-Scattering of Nanocrystals

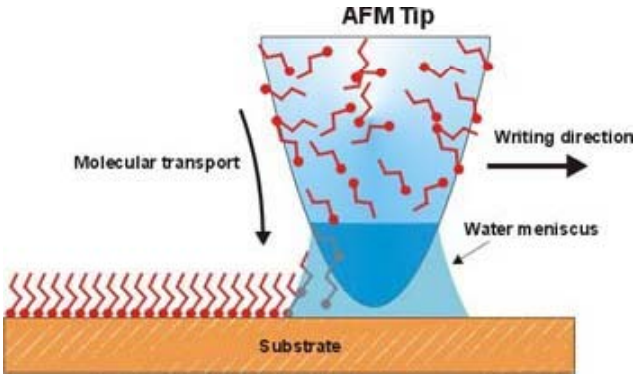


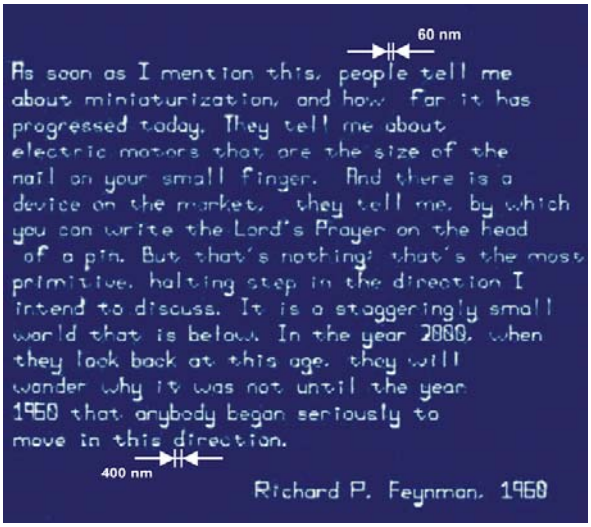
* The scale bar is the same for all the images.

Shape, Size, and Composition Matter!

Dip-Pen Nanolithography (DPN)



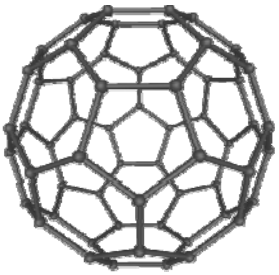






Chad Mirkin
(Northwestern University)

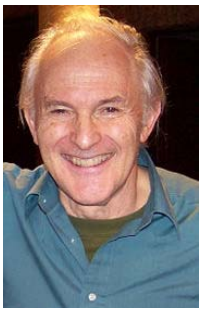
Carbon Allotrope - Buckminsterfullerene (C_{60})



Buckminsterfullerene (C_{60})
or bucykball



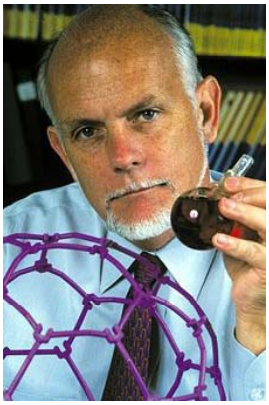
Geodesic dome by
R. Buckminster Fuller



Harold Kroto

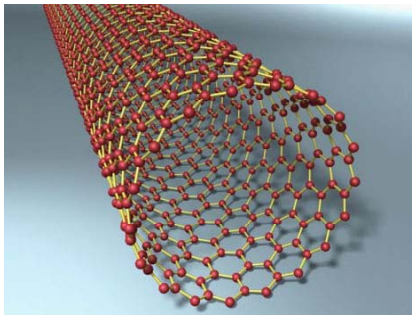


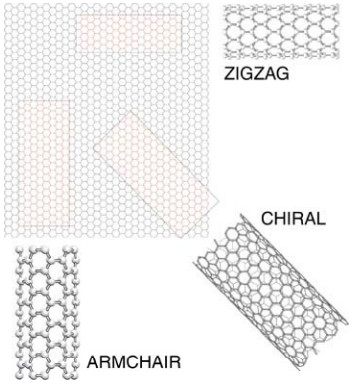
Robert Curl

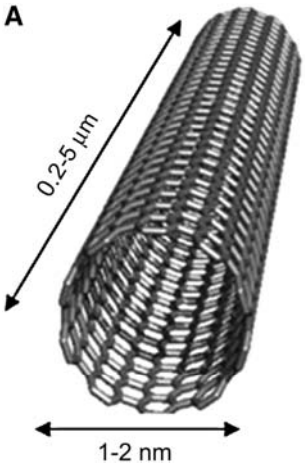


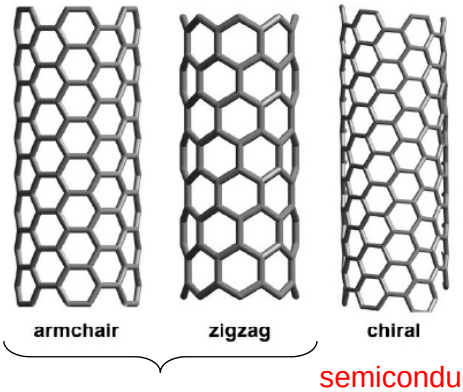
Richard Smalley

Carbon Allotrope - Carbon Nanotube







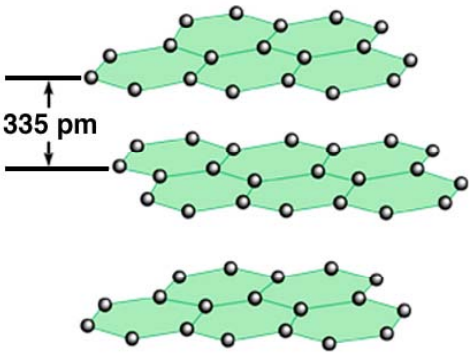


metallic, conductor

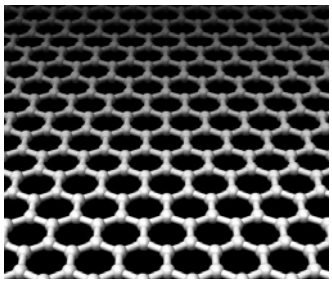
semiconductor

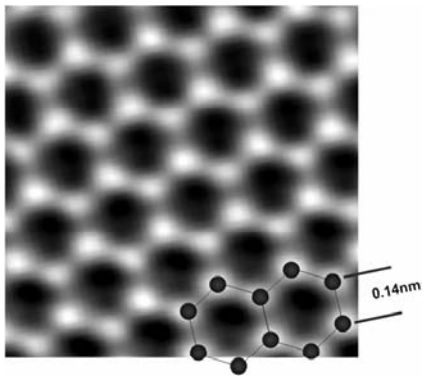
Carbon Allotrope - Graphene

- Graphite



- Graphene (Science **306**, 666 (2004))

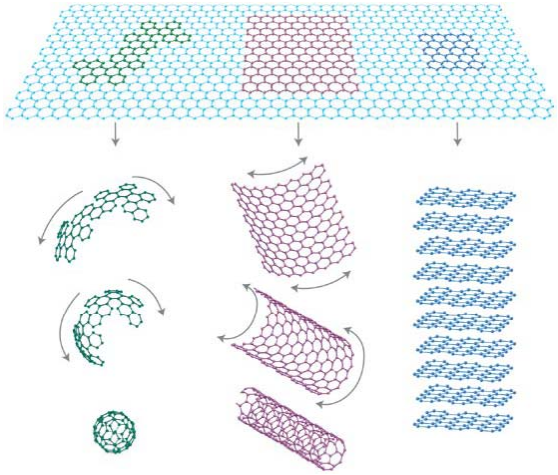




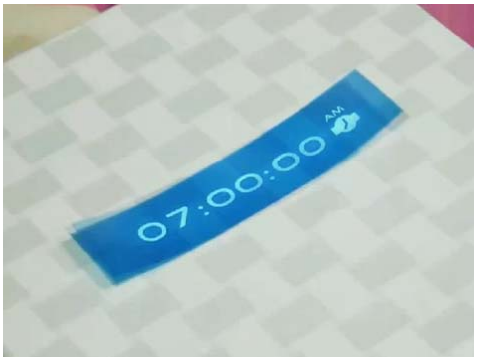
Transmission electron microscope (TEM) image of graphene

Graphene

- graphene – 모든 탄소 동소체의 모체
- graphene에 대한 가까운 미래의 응용



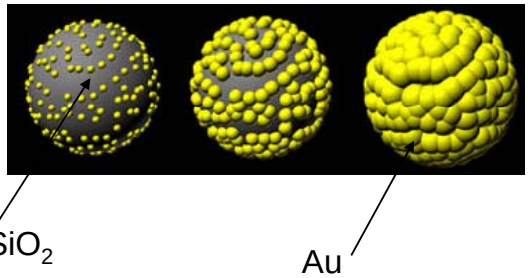
Geim, Nature Mater., **6**, 183 (2007)

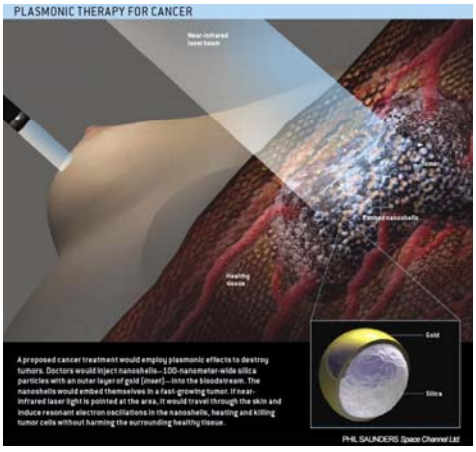


Nanoshells



Naomi Halas (Rice Univ.)

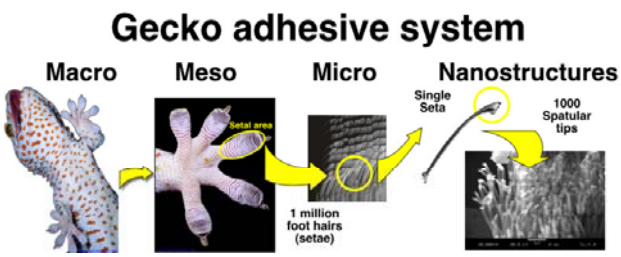


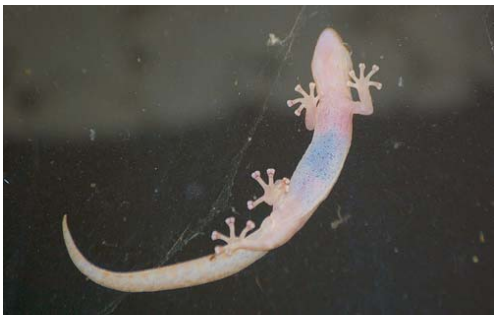


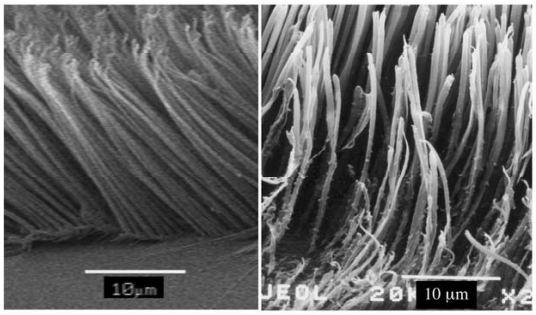
Nanoshells and breast cancer

Gecko and van der Waals Force

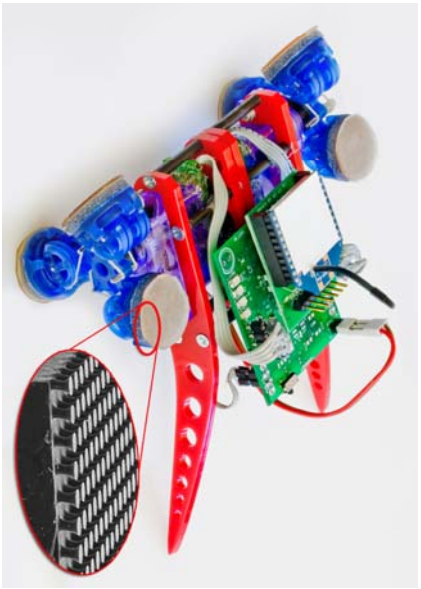






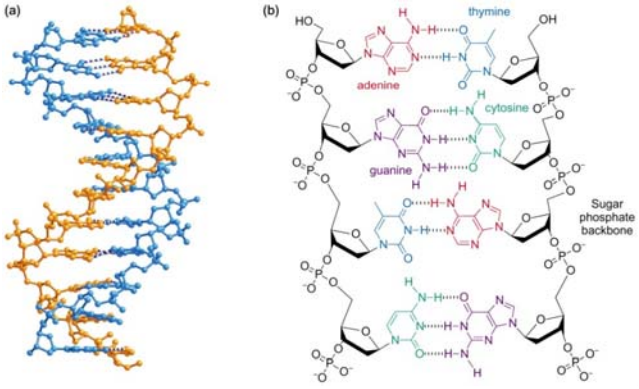


Gecko and van der Waals Force

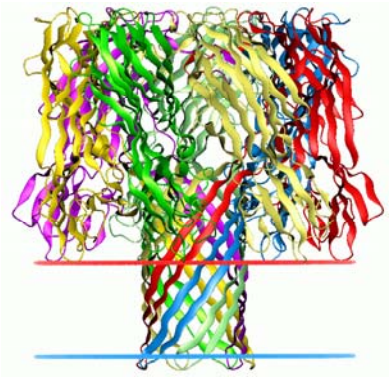




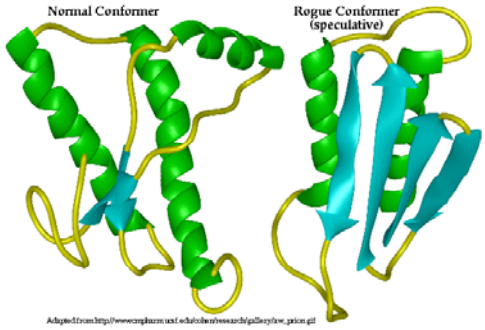
DNA and Proteins



DNA double helix

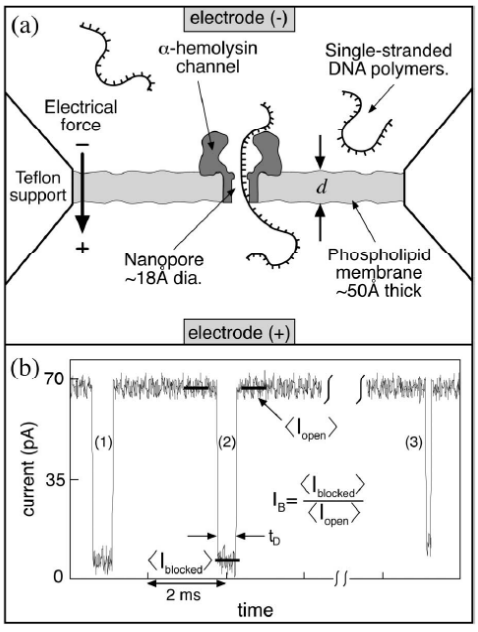


α-hemolysin

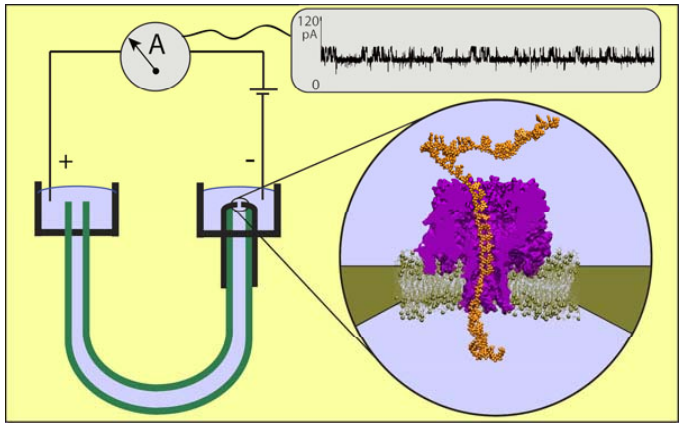


prion

DNA translocation

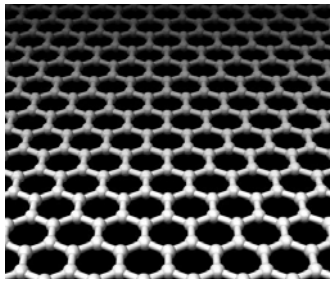


Miller et al., PRL **86**, 3435 (2001)

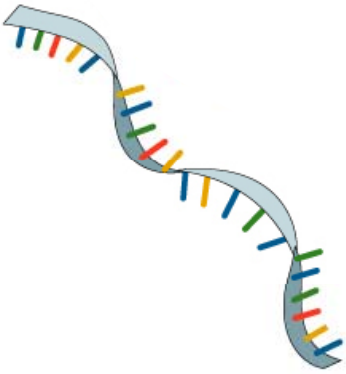


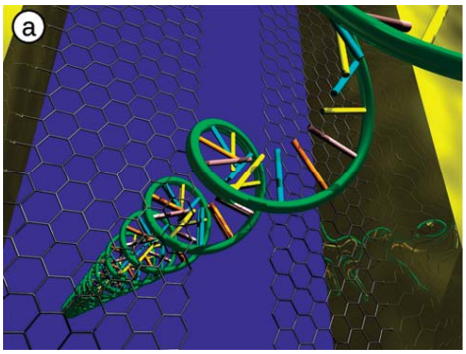
Schulten et al., PNAS **102**, 12377 (2005)

DNA translocation

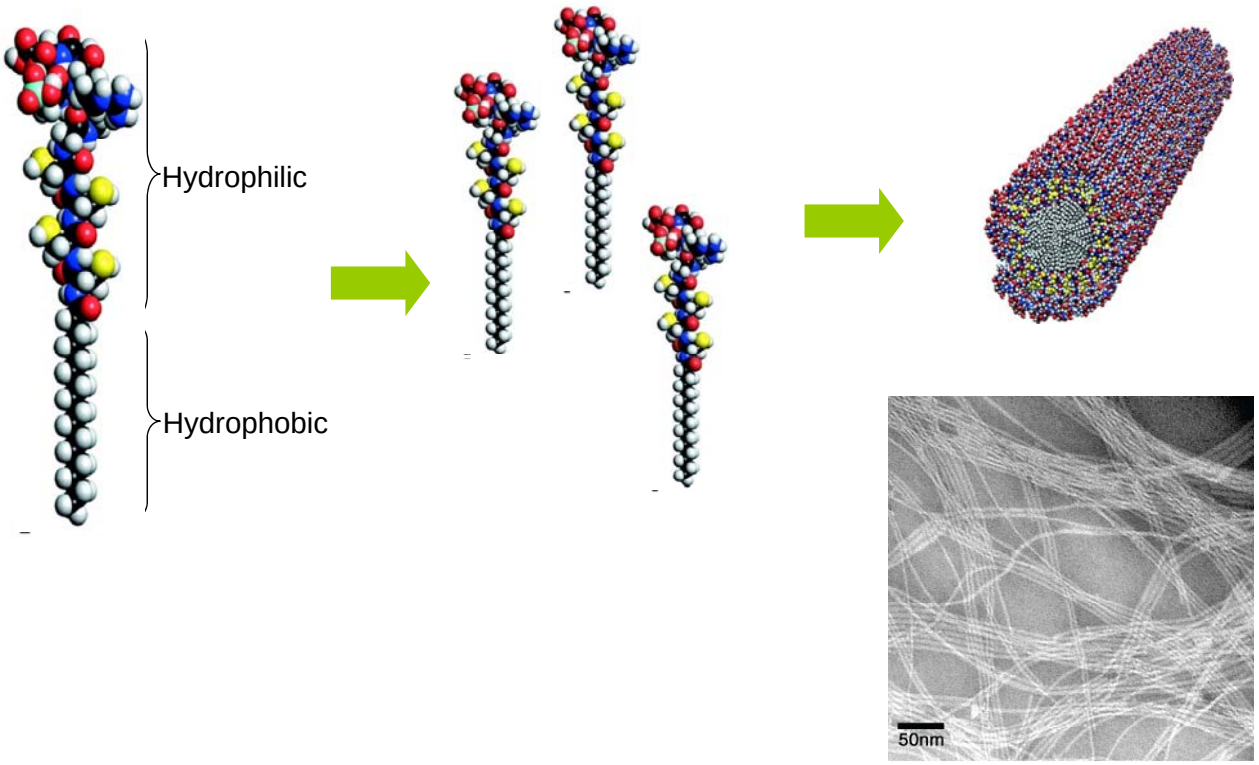


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Self-assembly of peptide amphiphile nanofibers



Hartgerink et al., Science **294**, 1684 (2001)